

Hardware Accelerator for PQC Algorithms

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Migration is no longer optional : the implementation deadline is set by federal mandate



2024

FIPS 203 / 204 / 205



2026

Executive Order + EU CRA



2030

PQC key establishment



2031

PQC digital signatures



FIPS 203/204/205 prove the algorithm, not the implementation



ML-KEM-1024



ML-DSA-44

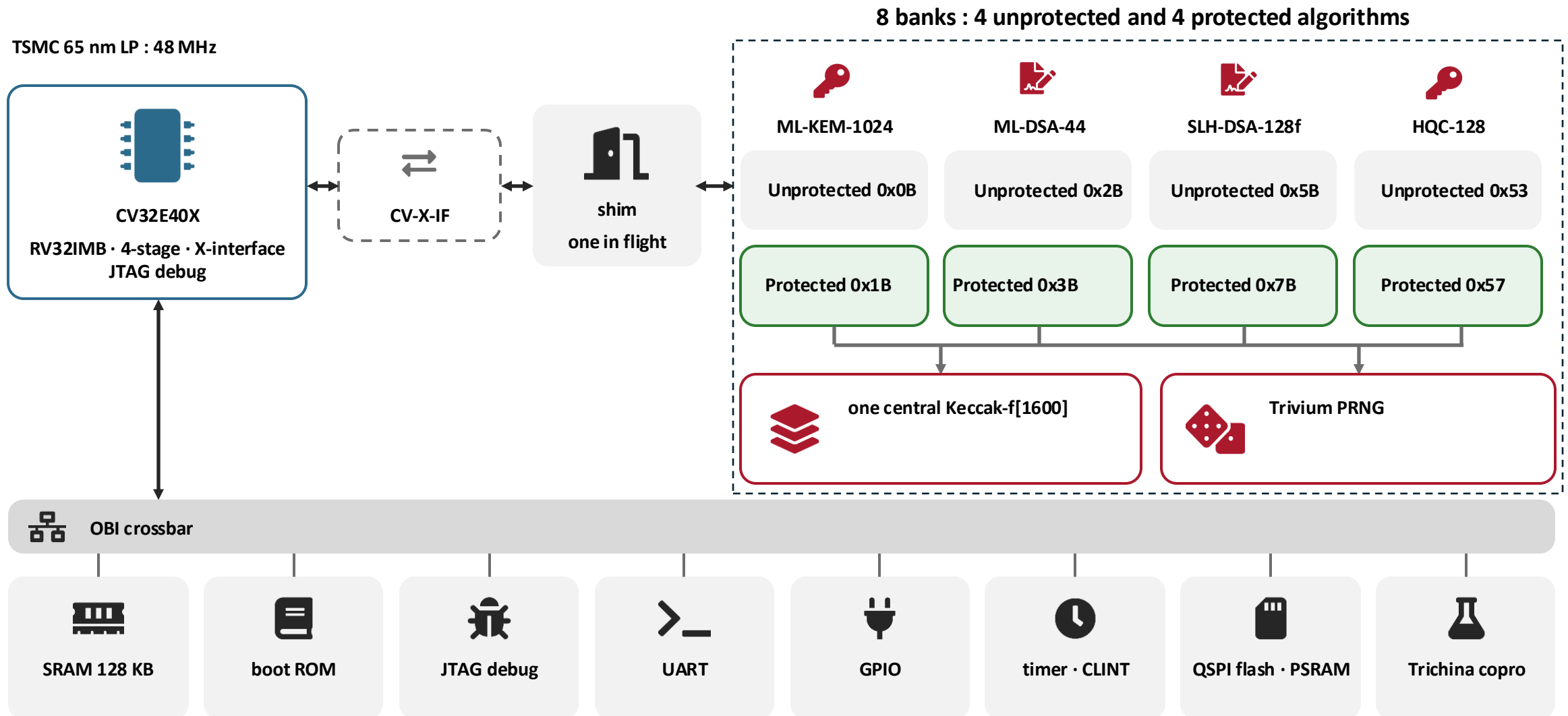


SLH-DSA-128f



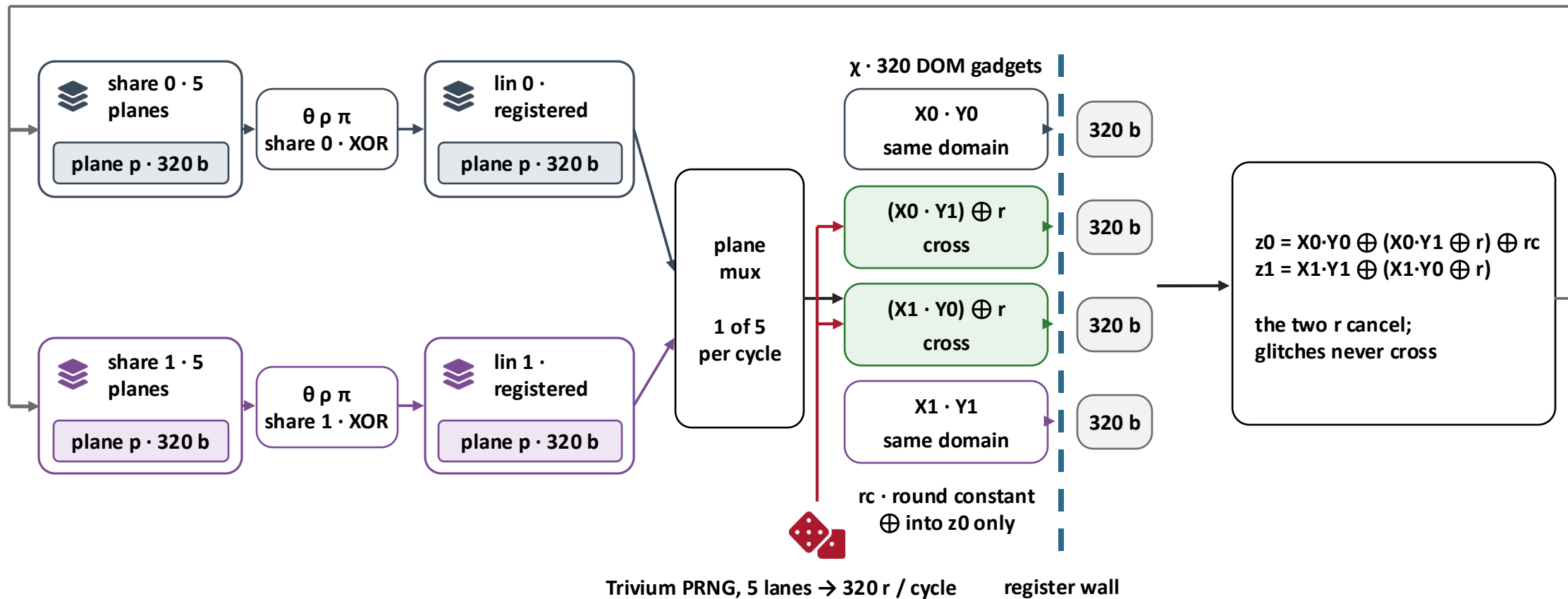
HQC-KEM-128

PRISM

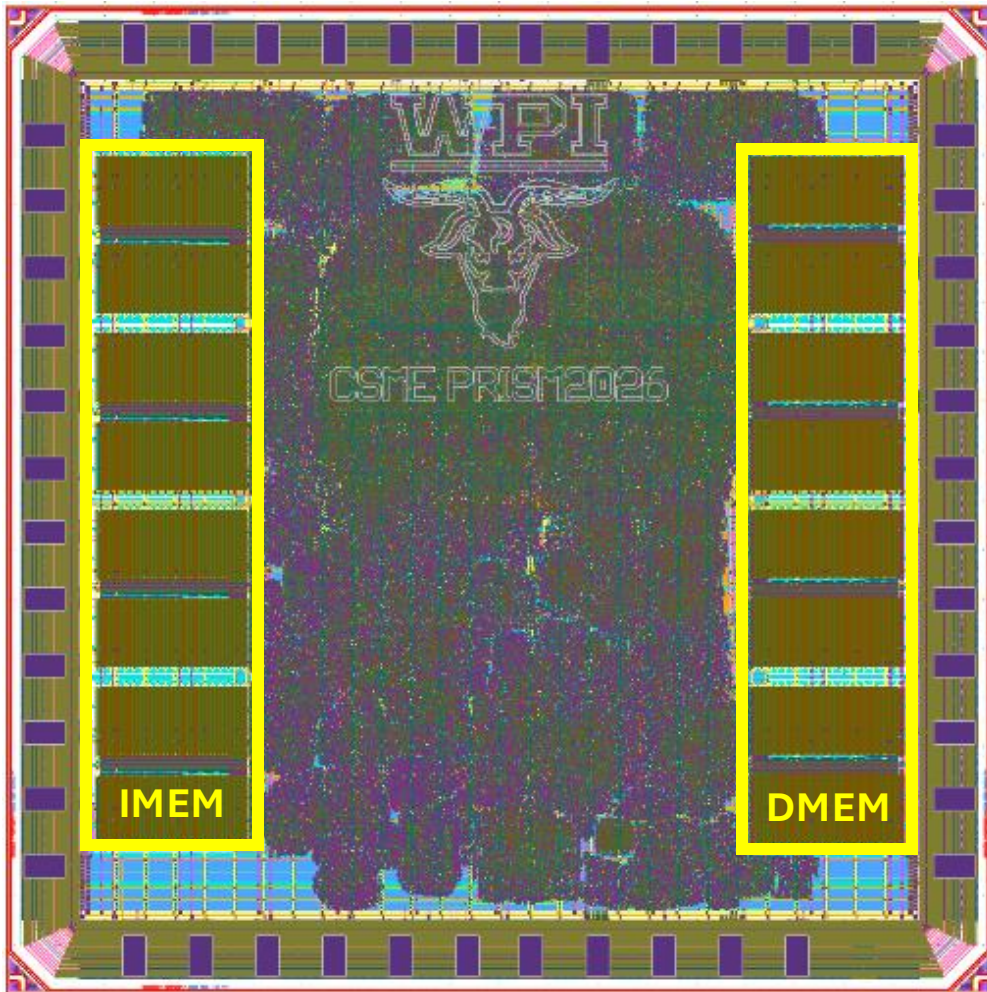


Keccak

write back: $z0 \rightarrow \text{share } 0$, $z1 \rightarrow \text{share } 1$



PRISM : Features



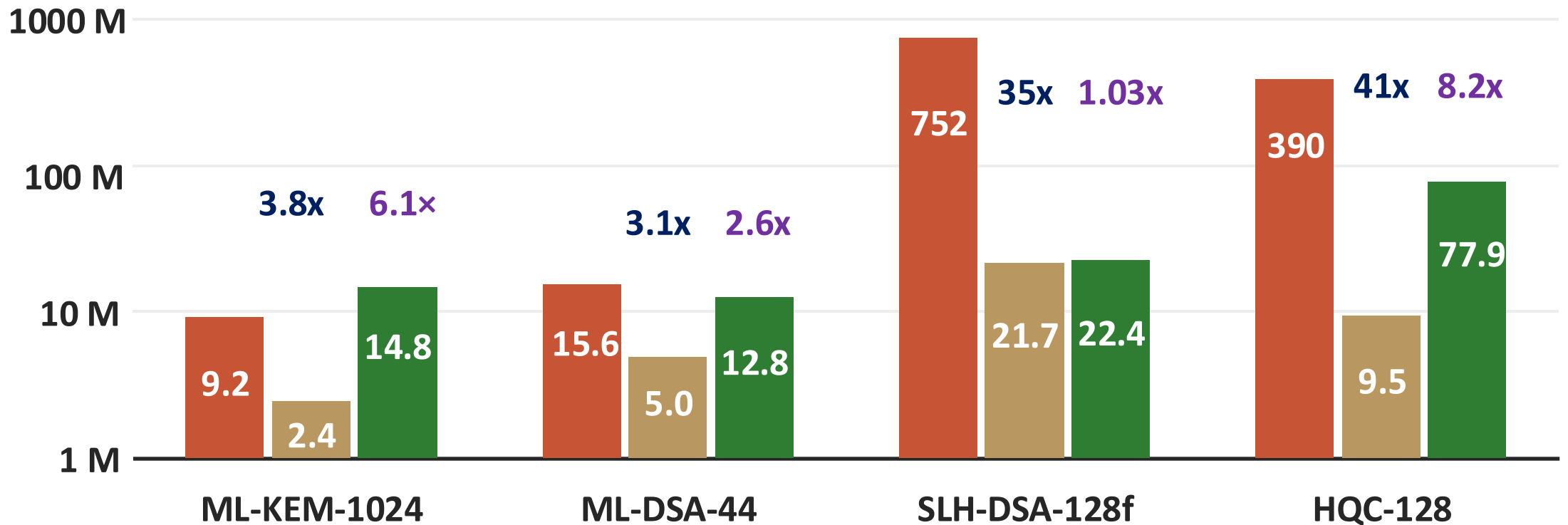
- Pre- and post-silicon SCA and fault platform
- unprotected and protected variants
- Observability: scan chain over all 30301 flops
- Share-0 and share-1 registers placed apart to reduce coupling
- 2 to 135 cycles per ISE instruction
- Unprotected ISE 3 to 41× faster than software on the same core

Performance

Cycles
(Millions)

software unprotected ISE protected ISE

$\text{speed-up} = \text{software} / \text{unprotected ISE}$ $\text{protection} = \text{protected ISE} / \text{unprotected ISE}$



Comparison

Features	HORCRUX	OpenTitan	PRISM
Algorithms	ML-KEM, ML-DSA, SLH-DSA, HQC	ML-KEM, ML-DSA, SLH-DSA(SW)	ML-KEM, ML-DSA, SLH-DSA, HQC
Countermeasure	None	Protected : MLKEM, MLDSA	Protected : ML-KEM, ML-DSA, SLH-DSA, HQC
ISE	32	Coprocessor firmware	85
Clock frequency	160 MHz : Synthesis	100 MHz	48 MHz
Speed-up	2 to 3x	Not reported	3 to 4x
Chip	Not fabricated	Earl Grey	PRISM
Pre-silicon SCA	No	formal masking checks	Simulation

Future plan

- Evaluate the protected coprocessors against side-channel attacks on the PRISM-65 chip
- Explore bit level physical effects

THANK YOU!